Enhancing Bipolar Disorder Diagnosis with Custom Loss XGBoost for Imbalanced Multi-Class Classification

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***Abstract*—Bipolar disorder is a challenging mental illness and is difficult to diagnose because of its complexity. This study uses machine learning to tackle the problem of imbalanced multi-class classification in bipolar disorder. A publicly available dataset from Harvard Dataverse was utilized, and an imbalanced dataset was created to evaluate performance under real-world class distributions. Entropy regularization, a novel custom loss function, and confidence penalty were incorporated into the XGBoost algorithm. An improvement of 14 percent in the F1-score was observed, indicating enhanced sensitivity to minority class detection and also the accuracy has improved by 4 percent. The new approach shows substantial improvement in the discovery of underrepresented classes like bipolar type - 1, indicating its possible application in clinical decision support systems for personalized psychiatric diagnosis.**

***Index Terms*—XGBoost, imbalanced classification, bipolar disorder, focal loss, entropy regularization, confidence penalty**

I. Introduction

Bipolar disorder or manic depressive illness is a serious mental health condition that involves fluctuating periods of mania, hypomania, and depression. It limits a person's cognitive, emotional and functional capacities which can affect social and occupational functioning of an individual. Bipolar disorder affects more than 1% of the world population and is connected with a reduced life expectancy around 10-20 years. Bipolar disorder is classified into bipolar type-1, bipolar type-2 and cyclothymic disorder based on the strength of high and low moods [11].

Bipolar type-1 disorder is defined by the presence of one or more manic episodes which may be preceded or succeeded by depressive episodes. Manic episodes involve elevated or irritable mood, increased energy, recklessness and cognitive impairments [11].

In contrast, bipolar type-2 disorder is characterized by recurrent episodes of depression along with hypomanic episodes, which are less intense than mania. The causes of bipolar disorder are multiple, and they include genetic susceptibility, neurobiological abnormalities, and environmental stressors [11], [13].

Effective planning of treatment and the overall quality of life in patients depends mainly on the accurate and timely diagnosis of the mental disorders. In most cases, it is made difficult due to the complexity of the mental condition. Specifically, the detection of less common yet clinically relevant diseases like bipolar type-1 is a particular challenge.

This study majorly focuses on this issue by observing the performance of XGBoost, one of the established strong and widely used gradient boosting algorithms [14], that can be used in situations related to imbalanced multiclass classifications for bipolar disorder diagnosis. We specifically focus on the issue of accurately detecting bipolar type-1. This is a condition that is relatively rare but has significant influence in a dataset where depression and bipolar type-2 are far more common. Recognizing that the standard loss function of XGBoost is not sufficient to accommodate imbalanced classifications, we developed a novel custom loss function that integrates focal loss, confidence penalties, and entropy regularization. This gives the model better sensitivity to minority-class instances so that bipolar type-1 can be more accurately classified.

Our results demonstrate that the custom loss function significantly improves the F1 score for bipolar type-1 and also increased the total accuracy of the XGBoost from 83% to 87% over the common objective. This study adds to the ongoing literature on creating more accurate and effective automated diagnostic instruments for mental health clinics, and ensuring that less prevalent conditions like bipolar type-1 are properly classified.

II. Related Works

Imbalanced data classification is a widely known problem in healthcare and clinical decision support systems. Various techniques have been introduced to handle class imbalance, including re-sampling, cost-sensitive learning and loss function adjustments like focal loss [1] - [3], [9].

Mental health contexts [4] mainly utilized support vector machines and random forests to classify bipolar disorder with clinical information, even though the models had difficulty with minority class performance. In a study, Sonkurt et al [5] utilized cognitive function tests and a machine learning model for separating bipolar disorder subtypes but cited difficulties based on biased class distribution. Another study by Rotenberg et al [6] investigated the prediction of relapse in bipolar disorder with supervised learning but described a lower sensitivity to identify manic states.

Our efforts incorporate a custom loss function that balances learning at the algorithmic level rather than solely through data-level techniques.

A. *Customized focal loss for imbalanced classification*

Imbalanced class distribution creates a severe issue in psychiatric datasets since some conditions such as schizophrenia or bipolar disorder occur far less frequently compared to more common disorders such as anxiety or depression. Conventional loss functions such as binary cross entropy tend to underperform in such cases because they do not adequately prioritize the minority classes. To fix this, researchers came up with focal loss [3], [9]. Here we employ a customized focal loss with entropy regularization and confidence penalty [7], [8], [10].

B. Entropy Regularization and Confidence Penalty

Entropy regularization and confidence penalty [7], [8], [10] methods were applied in machine learning to enhance model generalization and to avoid overconfident predictions. They are especially applicable to classification problems, where a model's confidence in its predictions can have a profound effect on performance and robustness.

C. Literature Shortcomings and Research Impact

The research on XGBoost, focal loss, entropy regularization, and confidence penalty [3], [7] - [10] have been conducted individually, their effect on minority classes has not been systematically studied.

This study will:

1) Increase the sensitivity of the XGBoost model to minority classes — particularly bipolar type-1 through adjusting the loss function.
2) Design a customized loss function that combines:
   - **Focal Loss** (to prioritize difficult-to-classify samples).
   - **Entropy Regularization** (to decrease overconfident predictions).
   - **Confidence Penalty** (to enhance generalization on minority classes).
3) Increase the F1-score and overall accuracy of the model.
4) Show that the new approach performs better than baseline XGBoost on an actual psychiatric dataset.
5) Better contribute to more accurate and fair automated diagnosis instruments for mental health clinics, particularly for less common but clinically meaningful conditions such as bipolar type-1 disorder.

III. Methodology

A. Dataset Description

Two datasets were utilized in this study. The primary dataset, `imbalance_500.csv`, contains 500 instances of patient data across four diagnostic categories labeled as *Expert Diagnose*.The class distribution was significantly imbalanced. Class 3 (n=220), Class 2 (n=200), Class 0 (n=50), and Class 1 (n=30). An auxiliary dataset, `Dataset_Mental_Disorders.csv` from Harvard Dataverse (which consisted of 120 patients) was used for initial feature alignment and preprocessing reference.The class distribution in this dataset was Class 3 (depression) n=31, Class 2 (bipolar type-2) n=31, Class 1 (bipolar type-1) n=28, Class 0(normal) n=30. But the modeling focused on the primary dataset which had 500 patients due to replicating the auxiliary dataset.

B. Preprocessing

All categorical variables were encoded using Label Encoder. The dataset was scaled using StandardScaler to normalize the feature space, ensuring that all features contributed equally to the model training process. The target variable (Expert Diagnose) was isolated for classification.

C. Data Splitting Strategy

The dataset was evenly stratified and split 50-50 into a training-validation subset and a testing subset. This was chosen not for maximizing training size, but to build a class-balanced test set that enables fairer per-class evaluation, particularly for underrepresented but clinically critical classes (e.g., Classes 0 and 1).

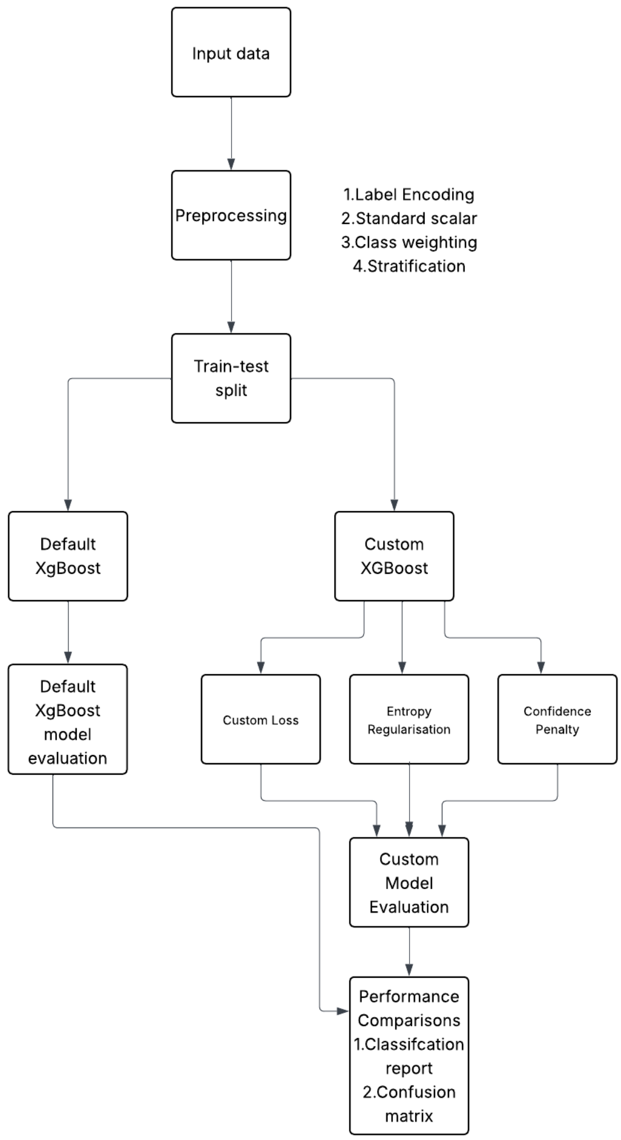

Fig. 1: Flowchart of the XGBoost Model Implementation

From the reserved 50% of the data for testing, a class-balanced test set was obtained by sampling, without replacement. Twenty-five data instances from each class were tested so that the test set equally represented the four classes: Class 0 (n=50), Class 1 (n=30), Class 2 (n=200), and Class 3 (n=220). Since each class had at least 25 samples, there was no concern for bias from sampling without replacement.

This test set design allowed the evaluation of precision, recall, and F1-score for each class to be positioned as independently as possible even when the classes were seriously imbalanced. A balanced test set is particularly critical in a clinical application, as the influence of performance on the lowest count class (class 1 with only 30 instances) can have disproportional implications for patient outcomes.

The other 50% of the data was then split 80/20 between training and validation, employing stratified sampling to maintain the original class distributions. It ensures evaluation fairness, without significantly compromising model performance, especially since class imbalance was explicitly addressed during training through class-weighted custom loss functions.

D. Class imbalance handling

Given the class imbalance in the training data, class weights were computed as:

$$w_i = \frac{N}{K \cdot n_i} \tag{1}$$

- w_i – The weight assigned to class i. Classes with fewer instances get higher weights to ensure the model pays more attention to them during training.
- N – The total number of training samples in the dataset. It represents the sum of all examples across all classes.
- K - The number of classes in the classification problem
- n_i – The number of training samples in class i.

E. Model and Loss Function

The flowchart of the XGBoost implementation is shown in Fig. 1.
The classification model was developed using the XGBoost framework with two configurations:

1) **Default XGBoost**, using the standard softmax loss (`multi:softprob`).
2) **Custom XGBoost**, using a novel objective function combining:
   a. **Focal Loss** – Down-weights easy samples and focuses learning on harder examples.
   b. **Entropy Regularization** – Encourages output distributions with higher entropy to prevent overconfident predictions.
   c. **Confidence Penalty** – Discourages the model from assigning excessively high probabilities to any single class.

The custom loss was defined as follows:

$$\text{Loss} = \alpha_t(1 - p_t)^\gamma - \lambda_H \cdot H(p) - \lambda_C \cdot \sum p^2 \tag{2}$$

- where p_t is the predicted probability of the true class.
- α_t is the class-specific scaling factor.
- γ is the focusing parameter.
- H(p) is the entropy of the predicted distribution.
- $\sum p^2$ is the the confidence penalty.

IV. Results and Discussion

A. Performance Metrics

The models are evaluated with the following Performance Metrics

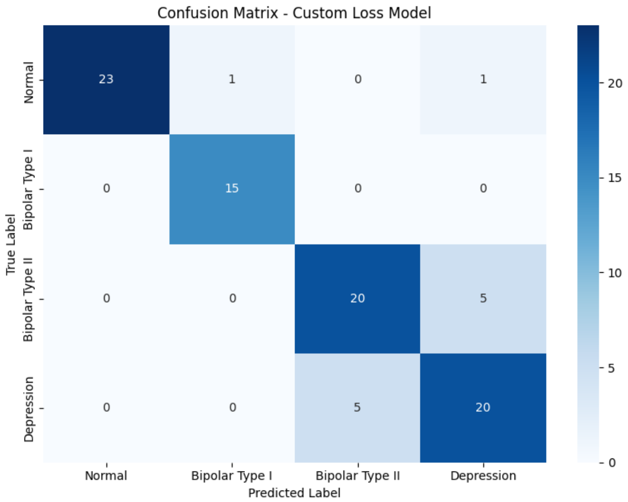

Fig. 2: Error matrix for the Custom Loss Model

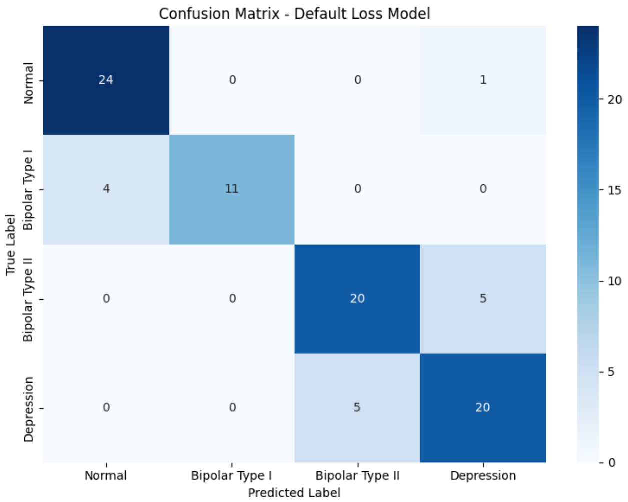

Fig. 3: Error matrix for the Default XGBoost

1) Accuracy
2) Precision
3) Recall
4) F1-score

1. Accuracy

Accuracy reflects the proportion of correct predictions made by your model.

2. Precision

Precision indicates the proportion of predicted positive cases that are truly positive.

3. Recall

Recall represents the proportion of actual positive cases that were correctly identified (TPR).

4. F1-Score

It harmonizes precision and recall by computing their harmonic mean.

TABLE I: Performance Evaluation of Custom XGBoost

Class	Precision	Recall	F1-Score	Support
0	1.00	0.92	0.96	25
1	0.94	1.00	0.97	15
2	0.80	0.80	0.80	25
3	0.77	0.80	0.78	25
Accuracy	-	-	**0.87**	**90**
Macro Avg	0.88	0.88	0.88	90
Weighted Avg	0.87	0.87	0.87	90

TABLE II: Performance Evaluation of Default XGBoost

Class	Precision	Recall	F1-Score	Support
0	0.86	0.96	0.91	25
1	1.00	0.73	0.85	15
2	0.80	0.80	0.80	25
3	0.77	0.80	0.78	25
Accuracy	-	-	**0.83**	**90**
Macro Avg	0.86	0.82	0.83	90
Weighted Avg	0.84	0.83	0.83	90

B. Error Matrix

An error matrix is an evaluation metric deployed in classification problems to summarize the output of a predictive model. It's a table with actual labels having a detailed presentation of correct versus incorrect classifications. The matrix consists of four elements: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). They help in assessing various performance indicators such as accuracy, precision, recall, and F1-score. From the error matrix, data scientists can determine the capability of the model.

C. XGBoost model Performance with custom loss

The model produces an accuracy of **87%.**

The performance evaluation of the XGBoost custom model is in Table I and the error matrix for the XGBoost custom model is shown in Fig. 2.

D. Default XGBoost model performance

The model produces an accuracy of **83%.**

The performance evaluation of the default loss model is in Table II and the error matrix for the XGBoost default model is shown in Fig. 3.

The custom loss model outperformed the default model, particularly in classes 0 and 1. This statement is supported by the evaluation metrics:

- **Class 0:** Higher F1-score (0.96 vs. 0.91), despite slightly lower recall.
- **Class 1:** Significantly higher F1-score (0.97 vs. 0.85), due to perfect recall and high precision.

E. Performance analysis of Standard models Vs XGBoost with Customloss function

The comparison of Performance analysis of Standard models Vs XGBoost with Customloss function is in Table III.

Class imbalance is effectively addressed by the focal loss-inspired formulation, which is responsible for the high precision, recall, and F1-scores obtained for classes 0 and 1 in the custom loss model. The focusing parameter (γ) downweights the contribution of well-classified examples and forces the model to focus on more difficult, incorrectly classified instances, particularly those from minority classes. Concurrently, the class-specific weighting term (α_t) increases the sensitivity of the model to underrepresented classes by amplifying their influence. This specific mechanism results in a significant improvement in Class 1 recall (from 0.73 to 1.00), and a slight decrease in precision (from 1.00 to 0.94) was observed, which raises the F1-score (from 0.85 to 0.97). By penalizing high-entropy outputs, the entropy regularization term $H(p)$ promotes confident but diverse classes, whereas the confidence penalty term $\sum p^2$ discourages overly confident misclassifications, promoting calibrated decision-making. Collectively, these components minimize both false positives by avoiding incorrect assignments with high certainty and false negatives by boosting true positive rates in minority classes.

TABLE III: Comparison of Standard Classifiers Vs XGBoost with Custom Loss

Limitation in Standard Classifiers	Advantages in XGBoost with Custom Loss
Focus on overall accuracy, not on minority classes	Uses focal loss to focus on minority class
Overfits small minority classes	Uses entropy regularization and confidence penalty and reduce overfitting in minority classes
Limited support for custom loss	Offers full flexibility in loss function design
Standard classifiers can use class weights to address class imbalance.But not effective in severe class imbalance	Provides more control(class specific-focussing) and better performance in multiclass settings

V. CONCLUSION AND FUTURE WORKS

Classic machine learning algorithms have a tendency to be biased toward the majority classes. The model's inability to correctly identify the minority classes is negatively impacted by this bias. Our framework effectively alleviates the difficulty imposed by class imbalance in the diagnosis of bipolar disorder. By utilizing a balanced test evaluation, stratified training and a novel loss function, we provided enhanced modeling support for fairness and interpretability.

This provides a potential avenue to applicability in a real-world clinical setting, in which reliable diagnoses of underrepresented classes are likely to occur. Future work may include integration of additional clinical features and collecting dataset from clinical institutions for better prediction accuracy of bipolar disorder.

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